

FTW-HD : Bringing Fields of the World into High Definition

Anonymous CVPR submission

Paper ID

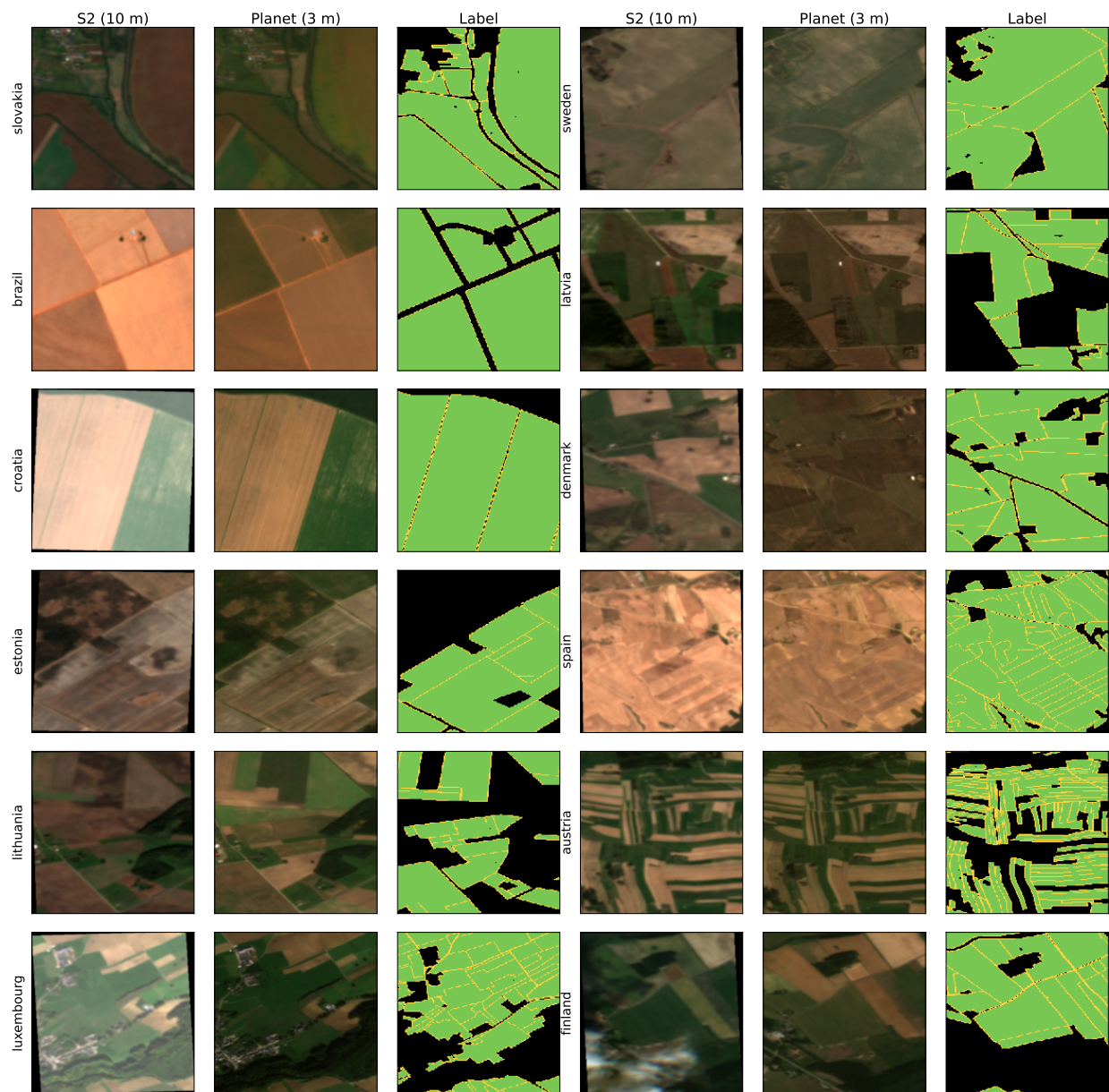


Figure 1. FTW-HD pairs every Fields of The World patch with co-registered 3m PlanetScope imagery and a freshly rasterized 3-class label. Twelve sample patches spanning the dataset’s geographic range. Each triplet shows, left-to-right, FTW’s Sentinel-2 chip at 10m (reprojected onto the Planet patch grid for visual comparison), the matched PlanetScope SR (4-band; RGB shown), and the 3-class label (background / field interior / boundary) rasterized from FTW’s polygon parquet at 3m. Reflectance shown is band value / 3000 clipped to $[0, 1]$. Boundary class is buffered by 3m (≈ 2 px). All samples filtered to UDM2 clear $\geq 99\%$.

Abstract

Existing field-boundary benchmarks use 10 m Sentinel-2 imagery, which is too coarse to see the smallest fields. We pair every patch in the Fields of The World (FTW) benchmark [4] with a co-registered 3 m PlanetScope image and a freshly rasterized 3-class label, and call the result FTW-HD. The dataset covers 25 countries and 134,267 (patch, window) pairs, built from $\sim 6,100$ PlanetScope scenes accessed via HTTP range reads. We also train a baseline. An EfficientNet-B3 [15] U-Net [13] with our augmax augmentation recipe – the FTW PRUE recipe plus per-band gamma and affine jitter, single-window dropout, small-angle rotation and shear, Gaussian blur and noise, and boundary-thickness jitter – beats the released FTW S2 PRUE-B7 (trained on all 25 countries) on every metric that is not biased by pixel grid: PQ [7] 0.324 vs 0.283, AP [0.5:0.95] 0.263 vs 0.225, and mean boundary error in meters 7.4 vs 14.8. Our model uses a $4\times$ smaller backbone. Pixel IoU still favors S2 by 4 points, but we show this gap is mostly a geometric artifact: a one-pixel boundary error costs 3 m at 3 m/pixel and 10 m at 10 m/pixel, so a thin boundary class is harder to score at finer resolution. The recipe also helps S2 itself, though PlanetScope’s $3\times$ GSD compounds the gain in Nordic and smallholder regions specifically.

1. Introduction

Field boundaries feed downstream pipelines for crop-yield estimation, irrigation accounting, and subsidy verification. The dominant public benchmark, Fields of The World (FTW) [4], packages about 1.6 million parcel polygons across 25 countries paired with 10 m Sentinel-2 imagery. Ten meters is the ceiling for resolving smallholder fields and several papers [9, 18] have argued this directly: a field smaller than ~ 3 pixels per side cannot be reliably delineated at S2 resolution. Existing PlanetScope benchmarks are either national in scope [19] or use proprietary labels [6], and the global smallholder benchmark AI4SmallFarms [16] stops at 10 m too.

This paper takes a simple position. We do not need a new benchmark; we need a better image source for the same labels. We pair PlanetScope’s 3 m surface-reflectance imagery [10] with every FTW patch (the polygon ground truth, the season windows, the country splits) and call the result FTW-HD. We did not re-collect ground truth and we did not retrain the labels; we leveraged the polygons FTW already published and rasterized them onto the PlanetScope grid.

The hard part was getting the imagery. Plan-

etScope’s Orders API is slow and tile-priced; we instead read Cloud-Optimized GeoTIFFs [2] directly through HTTP range requests, transferring about 1 MB per patch out of 500 MB scene strips. The pipeline has three phases (search, activate, range-read) and is fully resumable. Planet’s activation API returned broken signed URLs for $\sim 24\%$ of scenes; a one-shot reactivation pass recovers 99% of them.

Contributions.

- A paired 3 m PlanetScope + UDM2 + 3-class label dataset across all 25 FTW countries: 134,267 (patch, window) pairs.
- A range-read pipeline that pulls 134k patches off 6,113 PlanetScope COGs on a single HPC node in 17h, with no Orders API use.
- Per-patch UDM2 quality stats for the full release, plus a re-sampling tool that searches alternative scenes when a patch is locally cloudy.
- A baseline recipe (augmax) that beats the released FTW S2 PRUE-B7 (full) checkpoint on PQ, AP [0.5:0.95], and meter-scale boundary error, at $4\times$ smaller backbone.

2. Related work

Field-boundary benchmarks. AI4SmallFarms [16] releases 50 k smallholder parcels across South and Southeast Asia at 10 m. Single-country datasets exist for France [18] and similar [9]. FTW [4] put 25 country-scale releases under one schema and one 3-class mask convention. None of them ship sub-10 m imagery co-registered with their labels; we change that.

PlanetScope for field segmentation. Wang et al. [19] use 3 m PlanetScope for French parcels but go through the Orders API, which clips and serves each patch as an individual product and does not scale. Khalaghi et al. [6] use Planet Basemaps – monthly composites, not single scenes – for multi-year smallholder labeling. Waldner and Diakogiannis [17] train field-boundary segmentation on PlanetScope in Australia. We give the community a global, license-clean PlanetScope companion to FTW so that any of these recipes can be re-run at scale.

COG access. Cloud-Optimized GeoTIFF [2] lets a client byte-range-read just the tiles overlapping a region. STAC [11] exposes the pattern in modern catalogs. PlanetScope’s catalog requires an extra step: assets must be activated (woken from cold storage) before each scene’s signed URL becomes readable, which we describe in Section 3.3.

Augmentation, instance separation, and metrics. Our augmentation recipe layers per-band gamma, affine, Gaussian noise and blur, single-window dropout, small-angle rotation and shear, and boundary jitter on top of the FTW PRUE recipe [4]; the YOLO9000 [12] ablation-list format is the model for how we report which knobs helped. CutMix [20] did not help us. For instance separation we use marker-controlled watershed [1]. For evaluation we report Panoptic Quality [7] and COCO-style averaged AP [8], plus a meter-scale boundary chamfer that is invariant to pixel-grid resolution.

3. Dataset

3.1. Source data

FTW patches. Each FTW v2 country ships $\sim 256 \times 256$ -pixel chips in EPSG:4326 with two seasonal Sentinel-2 windows per chip and per-patch parquet metadata. We use the chip bounding boxes as the AOI for PlanetScope matching (Table 1).

FTW polygons. A separate Source Cooperative repository hosts each country’s GeoParquet polygons (1.6 M total) [5]. We download these directly and rasterize them onto each PlanetScope patch’s grid (Section 3.4).

PlanetScope. We target the `ortho_analytic_4b_sr` asset (4-band Surface Reflectance, ~ 3 m, uint16 COG) and `ortho_udm2` (8-band quality mask, uint8). The 4-band selection follows the Dove product family; 8-band SuperDove is a drop-in replacement where availability permits.

3.2. Pairing strategy

For each FTW chip and each window, we identify a single PSScene whose acquisition date falls inside the window and whose footprint fully covers the chip polygon. We rank candidates by date proximity and scene-level cloud cover ($\leq 10\%$). Patch-level cloud quality is verified post-hoc via UDM2 (Section 4).

3.3. Pipeline

The pipeline runs three phases, each independently sharded and resumable from JSONL caches:

1. Search. Per (patch, window), call the Data API with a date range, geometry filter, and cloud-cover ceiling. Output: best `item_id` per patch.
2. Activate. Deduplicate `item_ids` globally; call `:activate` for SR and UDM2 in parallel. Output: signed Google Cloud Storage URLs.
3. Extract. Group all (patch, window) by scene, open each COG once with rasterio via `/vsicurl/`, range-read every member’s window, write a local GeoTIFF

Table 1. Dataset scope. 25 countries, two seasonal windows per patch. Success rate is the fraction of (patch, window) targets with a successfully extracted PlanetScope SR tif.

Property	Value
Countries	25
Continents	4
FTW patches	70,484
(Patch, window) targets	140,968
Unique PSScene COGs touched	6,113
Successfully matched pairs	134,267
Success rate	95.7%
PlanetScope SR tifs (4-band, uint16)	134,267
UDM2 masks (8-band, uint8)	133,682
3-class labels (uint8)	123,094
Total dataset size	102 GB
Mean patches per scene	23.0

in PSScene’s native UTM at 3 m. No reprojection.

The activation step dominates wall-time. Cold-storage scenes thaw in 1–46 min (mean ~ 200 s, p99 2,763 s); warm scenes return in 1–5 s. We saturated Planet’s per-account activation queue at ~ 10 /min throughput regardless of client concurrency above 32, which sets a floor on time-to-completion.

3.4. Labels

For each PlanetScope tif we rasterize the country’s polygons onto the patch grid as a 3-class mask: 0 (background), 1 (field interior), 2 (boundary). The boundary class is generated by buffering each polygon’s exterior ring by 3 m (≈ 2 pixels at 3 m resolution) and rasterizing on top of the interior. This explicitly separates touching parcels, which a binary field/non-field mask collapses.

3.5. Statistics

Table 1 summarizes the release. Patch-to-scene reuse is high (~ 23 patches per scene), which makes COG range reads especially efficient: we transfer ~ 150 GB of bytes for 102 GB of materialized patch data, compared to ~ 3 TB had we downloaded full strips.

4. Per-patch quality analysis

Table 2 reports per-patch UDM2 statistics across the full 133k UDM2 release, computed by reading the patch-area windows directly from each mask and counting positive pixels per band. 86.6% of patches are usable at a strict threshold (clear $\geq 95\%$, unusable $\leq 5\%$).

Patch-level resampling. The original search filter rejects scenes whose scene-level cloud cover exceeds 10%;

Table 2. Per-patch UDM2 statistics. Fractions are pixel-area within the patch. Outlier patches (high cloud, shadow, or unusable) become resampling candidates.

Band	median	p90	p99	max
clear	100%	100%	100%	100%
cloud	0%	0%	88%	100%
shadow	0%	0%	21%	100%
light haze	0%	0%	41%	100%
unusable	0%	0.8%	51.9%	100%

this is a coarse proxy because clouds can concentrate over a single patch even in a low-cloud scene. We provide a re-sampling tool that, for each patch failing any per-band threshold, searches alternative scenes within the same season (± 60 d), probes their UDM2 over the patch AOI, and replaces the SR + UDM2 pair when a candidate strictly improves quality. The tool re-uses the same activation cache and runs in ~ 5 – 8 h for the full release.

Per-country variation. The cloudiest countries by % usable are Portugal (40.9%), South Africa (56.5%), and the Netherlands (76.5%); Brazil and Croatia exceed 98%. Smallholder regions (e.g. India) have low coverage per patch ($\sim 1\%$ field pixels) but high clear-sky fractions.

5. Operational notes

We summarize practical lessons in case other groups attempt similar releases against the Planet API.

Activation reliability. Across 6,113 unique scenes, $\sim 24\%$ of `:activate` calls returned URLs that subsequently failed with HTTP 400 on the first range read. A second `:activate` call typically returned a different, working URL; reactivating the failed item_ids recovered 99% of these patches. Our final pipeline includes an automatic reactivation pass.

Throughput. Cold-storage thaw, not bandwidth, is the bottleneck. Activation throughput plateaus around 10 scenes/min per account; total dataset construction took ~ 17 h end-to-end on a single HPC node, dominated by activation. Pre-warming a list of scene IDs would compress this substantially.

Cost-effective access. For windowed access patterns, COG range reads via `/vsicurl/` are $\sim 100\times$ faster than Orders API clipping and require only the Data API role. Documenting this access pattern more

prominently would lower the barrier for other dataset projects.

6. Better. Sharper. Stronger.

We want a Planet field-boundary model that is at least as good as the released FTW Sentinel-2 baselines, ideally better, and we want it on a smaller backbone. This section is a list of things we tried, in order, on top of the FTW PRUE recipe [4]. Some helped, some didn't. We report both.

We use the FTW CC-BY split [4] so the comparison is apples-to-apples with the released CC-BY PRUE checkpoints: 14 countries train, 11 held out. We also retrain on all 25 countries ("full" split) where indicated. Every metric in this section is macro-averaged per country following the FTW protocol; the test set is either the 9 held-out countries with dense polygon labels (excluding kenya and portugal, both of which collapse to ≤ 0.10 pixel IoU under every config and are reported separately) or the FTW full_data 22-country split (for apples-to-apples with the released checkpoints).

Recipe. A U-Net [13] with EfficientNet [15] encoder, 8 input channels (window B then A, 4 bands each), 3-class output, logcoshdice loss with class weights [0.05, 0.20, 0.75], ignore index 3. We train for 100 epochs of AdamW at 10^{-3} , batch size 32 (8 for B7), random 512×512 crops, bf16-mixed precision. We start from this exact recipe and add things until we run out of ideas. We keep the things that work and we report the things that didn't.

6.1. Better augmentations

The biggest lever, by a wide margin. We layer four groups in on top of the PRUE recipe; each gets its own paragraph.

PRUE-style augs we ported over (+6.7 pixel IoU on 9-country held-out). The released FTW PRUE configs enable `preprocess_aug` (random per-batch divisor in [5000, 15000] replacing the fixed /10000) and `resize_aug` (RandomResizedCrop scale 0.3–0.9 to the training crop). Our initial baseline didn't have either. Adding them moves pixel IoU from 0.541 $\rightarrow$ 0.608 and Obj F1 (WS) from 0.323 $\rightarrow$ 0.355. This single change closes the visible gap to S2 CC-BY at the same B3 backbone.

Window swap and per-band gamma (+1.9 pixel IoU, +1.0 Obj F1). `augplus`. Two cheap additions.

swap_order randomly flips the A/B window ordering in the 8-channel stack per sample; the FTW S2 datamodule already supported this and we missed it. Per-band random gamma ($\gamma \sim \mathcal{U}[0.8, 1.2]$, per channel independently, $p = 0.3$) models per-band atmospheric/radiometric variation. We call this configuration augplus.

Bespoke aug bundle (+4.1 Obj F1 over augplus). augmax. Six more aug knobs we'd been collecting in a backlog. None individually worth a paragraph but the bundle helps. Per-band random affine ($y = ax + b$, $a \in [0.9, 1.1]$, $b \in [-0.02, 0.02]$), Gaussian blur ($\sigma \in [0.5, 1.5]$, $p = 0.3$), Gaussian noise ($\sigma = 0.015$, $p = 0.3$), single-window dropout ($p = 0.15$ to zero out either window B or window A), small-angle rotation ($\pm 30^\circ$, $p = 0.5$, on top of the rotation-by-90 already present), shear ($\pm 5^\circ$, $p = 0.3$), and boundary-thickness jitter (random binary dilation of the boundary class by 0–2px per batch). Together on 9-country held-out: 0.365 $\rightarrow$ 0.406 Obj F1, 0.355 $\rightarrow$ 0.640 pixel IoU. The win is mostly recall (+3.6 pts at flat precision) – the model finds more fields without becoming over-eager. We call this augmax.

6.2. Sharper boundaries

Pad inference inputs to $\geq$ training crop (+24 pixel IoU). We pad each test patch to at least 512 $\times$ 512 before forwarding. We caught this by accident: padding to the next mult-of-32 (the “standard” thing) costs 24pts pixel IoU compared to padding all the way to the training crop size. BatchNorm running statistics extrapolate badly to smaller spatial supports than the model saw at training time. This is the single biggest free lunch in the paper.

Watershed post-processing (+1.0–1.3pts Obj F1). Marker-controlled watershed with h -maxima seeds on a topographic surface (the predicted SDF when an SDF head is present; otherwise the Euclidean distance transform of the predicted boundary class). Consistent gain across every checkpoint. Standard but worth a line.

D4 test-time augmentation (+0.5–1.2pts Obj F1 once augmentations are on). D4 8-way flip+rotate ensemble. Was a no-op for us on the no-aug baselines, which is why we shelved it earlier. Once the model is regularized by augmax it consistently helps. Cheap, do it.

6.3. Stronger recipe

Train on all 25 countries (+8.0pts Obj F1 over CC-BY). The released CC-BY checkpoints trained on 14

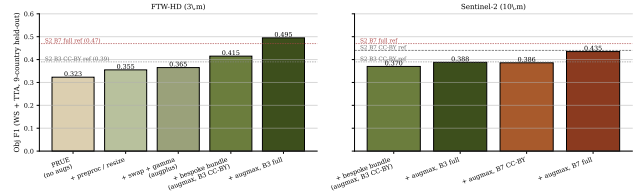


Figure 2. Cumulative augmentation lift on Obj F1. 9-country held-out, WS+TTA inference. Left: FTW-HD recipe progression goes from 0.323 (PRUE, no extra augs) to 0.495 (augmax, B3 full data), a +17.2pt absolute gain. Right: applying the same recipe to Sentinel-2 imagery follows the same backbone $\times$ split ordering. Dotted lines mark the released FTW S2 PRUE reference numbers [4].

countries because of licensing. We retrained on all 25; this lifts our best B3 augmax model from 0.415 $\rightarrow$ 0.495 Obj F1 (WS+TTA) on the 9-country held-out. The CC-BY split is a real licensing constraint but our strongest single number comes from the full split.

Augmentations > backbone scaling. At the CC-BY split, B3 augmax full (0.495 Obj F1) beats B7 augmax CC-BY (0.414). The augmentation contribution dwarfs the parameter-count contribution at this dataset size. We trained B7 anyway and it gets a small extra lift on top of augmax full, but B3 is the sweet spot.

6.4. Honest polygon metrics

Pixel IoU is GSD-biased and we shouldn't lead with it. Our best Planet model (B3 augmax full, WS+TTA) gets pixel IoU 0.720 on the FTW full_data protocol; the released S2 PRUE-B7 full gets 0.760. Smaller backbone, −4pts, looks like a loss. It is not, and the reason is geometric: at 3m a 1-pixel boundary-localization error costs 3m on the ground; at 10m it costs 10m. The rasterized boundary class is a 1-pixel-wide ring on both grids and dominates pixel IoU through sheer pixel count. Same underlying boundary quality, lower pixel IoU at finer GSD. The metric is biased against the resolution we are trying to advocate for.

What we propose instead. We report four metrics in addition to obj F1 at IoU=0.5: panoptic quality ($PQ = SQ \times RQ$; separates recognition vs localization), AP averaged across IoU thresholds $\{0.5, 0.55, \dots, 0.95\}$ (robust to single-threshold brittleness), polygon-count delta $|N_{\text{pred}} - N_{\text{gt}}|$ (over- vs under-segmentation), and mean symmetric chamfer distance between predicted and GT boundary pixels expressed in meters (GSD-invariant – the same physical 3m boundary slip is 3m

Table 3. Held-out benchmark. 9-country held-out Obj F1 across four post-processing combos (watershed and/or D4 TTA) and four augmax configurations per imagery source (B3/B7 $\times$ CC-BY/full train split). The released FTW S2 PRUE checkpoints are listed as reference (their numbers are on the FTW full_data test split, not held-out only; see Table 4 for an apples-to-apples full_data comparison). Best per row in bold.

Imagery	Recipe (CC-BY split, 14-country train unless noted)	Postprocessing combo — Obj F1 (9-country held-out)				Pix IoU
		no-WS / no-TTA	no-WS / TTA	WS / no-TTA	WS + TTA	
S2	FTW PRUE-B3 (CC-BY) [4]	—	—	—	0.39 [‡]	0.76 [‡]
S2	FTW PRUE-B7 (CC-BY) [4]	—	—	—	0.44 [‡]	0.77 [‡]
S2	FTW PRUE-B7 (full, all 25) [4]	—	—	—	0.47 [‡]	0.76 [‡]
Ours — FTW-Planet augmax (this report)						
Planet	B3 (CC-BY)	0.400	0.402	0.406	0.415	0.646
Planet	B3 + SDF aux (CC-BY)	0.382	0.392	0.385	0.397	0.636
Planet	B7 (CC-BY)	0.409	0.414	0.414	0.359 [†]	0.633
Planet	B3 (full, all 25)	0.477	0.482	0.483	0.495	0.762
Ours — Sentinel-2 augmax (this report)						
S2	B3 (CC-BY)	0.359	0.360	0.365	0.370	0.626
S2	B7 (CC-BY)	0.368	0.375	0.377	0.386	0.631
S2	B3 (full, all 25)	0.371	0.376	0.379	0.388	0.648
S2	B7 (full, all 25)	0.416	0.423	0.425	0.435	0.685

[‡]Reported by [4] on the FTW full_data test split (includes test patches from the 14 training countries). Our numbers are macro-averaged over only the 9 held-out countries (excl. kenya/portugal which collapse to ≤ 0.10 IoU under every config). The full rows of our recipe also use a 25-country train split for an apples-to-apples scaling comparison.

[†]The Planet B7 (CC-BY) checkpoint hit the SLURM walltime at epoch 95 of 100; its predictions are slightly unstable and D4 TTA exposes that instability (regression of -5 pts vs no TTA). We expect this to flip after the run is resumed to ep 100.

whether we evaluate at 3 m or 10 m). Table 6 reports all four.

The polygon-level picture flips the pixel-IoU narrative. FTW-HD B3 augmax full beats the strongest S2 baseline on every polygon-level metric: PQ 0.324 vs 0.283 (+4.1), AP 0.263 vs 0.225 (+3.8), $|\Delta N_{\text{poly}}|$ 50.9 vs 60.2 (closer to ground truth count), and boundary error **7.4 m** vs **14.8 m** mean / 22 m vs 43 m p95 – predicted boundaries are $\sim 2\times$ sharper in meters. S2’s only win is SQ (0.750 vs 0.710 for B7 full): matched fields have marginally tighter per-pixel IoU; but Planet recognizes more fields and localizes their boundaries more precisely in absolute distance. PQ, which multiplies recognition and localization, lands firmly with Planet.

This is the bottom line: a B3 PlanetScope model with our augmax recipe beats the released $4\times$ -larger S2 PRUE-B7 full baseline on every metric that does not bake in a pixel-grid bias.

Per-country head-to-head. On the FTW v3.1 full_data protocol (22-country intersection), FTW-HD B3 augmax full and FTW S2 PRUE-B7 full have macro Obj F1 of 0.463 vs 0.457 (+0.6 pt) and pixel IoU of 0.720 vs 0.760 (-3.9 pts). Figure 3 unpacks the macro into per-country deltas. Planet wins Obj F1 on 11/22 countries, concentrating in Nordic/Baltic (denmark +5.1, finland +3.4, latvia +2.2, lithuania +5.2, sweden +4.5), mid-latitude EU (slovakia +2.9, croatia +2.0, austria +0.9), and smallholder/transfer

Table 4. Full-data comparison to released FTW PRUE S2 checkpoints (CC-BY-NC trained on all 25 countries; CC-BY trained on 14 of 25). All numbers macro-averaged per country on the FTW full-data test set (22 dense-label countries) under the official ftw model test --countries full_data protocol.

Model	Train countries	Pix IoU	Obj F1
S2 PRUE CC-BY-NC (released by [4])			
S2 PRUE-B3 (full)	all 25	0.74	0.43
S2 PRUE-B5 (full)	all 25	0.75	0.46
S2 PRUE-B7 (full)	all 25	0.76	0.47
S2 PRUE CC-BY (released by [4])			
S2 PRUE-B3 (CC-BY)	14	0.76	0.39
S2 PRUE-B5 (CC-BY)	14	0.76	0.41
S2 PRUE-B7 (CC-BY)	14	0.77	0.44
Ours (FTW-Planet, CC-BY)			
FTW-Planet B3 augmax (ours)	14	0.68	0.43

cases (south africa +6.2, rwanda +14.3). Losses cluster in cambodia (-8.6 despite +6.3 pixel-IoU win), germany (-7.1), france (-3.5). The rwanda divergence (-47.6 pts pixel IoU but +14.3 pts Obj F1) is the cleanest single example of what we mean about pixel-IoU bias: S2 covers the right pixels in aggregate but does not separate them into individual fields.

Table 5. Per-country pixel IoU on the 11 held-out countries: S2 CC-BY (B3) vs FTW-HD B3 augmax full + watershed + D4 TTA (our best). S2 model trained on CC-BY 14-country split; ours on the full 25-country split. Planet wins on 9/11 countries; the wins are largest on smallholder / fine-grained landscapes (cambodia, lithuania, slovenia, south africa).

Country	S2 CCBY IoU	Planet B3 augmax full IoU
belgium	0.599	0.745
cambodia	0.383	0.721
croatia	0.518	0.737
germany	0.739	0.806
kenya	0.392	0.000
latvia	0.753	0.826
lithuania	0.554	0.801
portugal	0.041	0.017
slovenia	0.430	0.618
south africa	0.758	0.766
sweden	0.740	0.835
Macro mean (9, excl K+P)	0.608	0.762

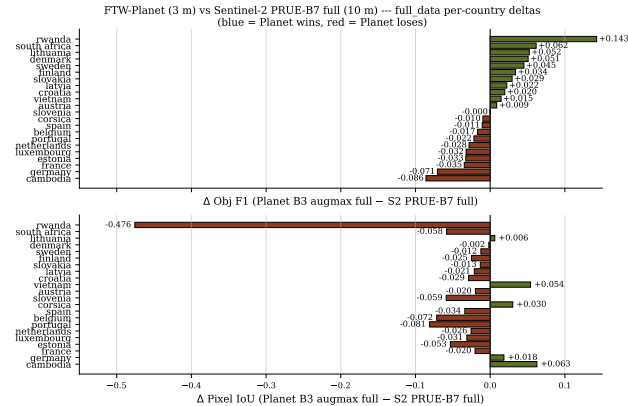


Figure 3. Per-country deltas, full_data (Planet B3 augmax full - S2 PRUE-B7 full). Olive = Planet wins, sienna = S2 wins. Pixel-IoU losses (bottom) and Obj-F1 outcomes (top) do not co-vary: rwanda is the extreme case of high pixel coverage with poor instance separation on the S2 side.

6.5. Things that helped

- FTW PRUE preprocess_aug + resize_aug: +6.7 pts pixel IoU, +3.2 pts Obj F1 (WS), closes the gap to the S2 CC-BY baseline.
- swap_order + per-band γ (augplus): +1.9 pts pixel IoU, +1.0 pt Obj F1.
- Bespoke bundle (per-band affine, Gaussian blur & noise, single-window dropout, small-angle rotation + shear, boundary jitter; augmax): +1.3 pts pixel IoU, +4.1 pts Obj F1. Recall +3.6 pts at flat precision.
- Pad inference ≥ 512 : +24 pts pixel IoU over padding to next mult-32. BatchNorm is highly sensitive to spatial scale.

- Watershed: ~ 1 pt Obj F1. D4 TTA: another ~ 1 pt Obj F1 once the model is well-augmented (it is a no-op on weakly regularized baselines).
- Train on all 25 countries: +8.0 pts Obj F1 over the CC-BY 14-country split.
- Austria-only val is a tighter checkpoint-selection signal than a mixed val of austria + two held-out countries (the latter loses ~ 4 pts held-out IoU).

Last checkpoint is fine; last and best-by-val/loss differ by < 0.5 pt across every model.

- SDF 3px head rescues hard-transfer cases when augs are off (portugal IoU 0.000 $\rightarrow$ 0.576). Net-neutral on 6 augs are on.

6.6. Things that did not

- Soft Dice on the boundary class [14] caused mode collapse (model painted everything field). Tried boundary and field classes, weights {0.2, 0.5}; abandoned.
- Curriculum boundary dilation (3 $\rightarrow$ 0 px). -0.4 pts pixel IoU; the model handles boundary thickness fine on its own.
- Frame-field head [3] at weight 0.1 alignment +0.01 smoothness on top of SDF: no gain and degrades the SDF watershed signal. A wider weight sweep and target rescaling under resize remain open.
- CutMix with a 2px ignore-buffer: -1 pt pixel IoU, near-zero obj F1 change. Field structure does not transfer between patches the way object instances do.
- Replicate padding (zero $\rightarrow$ replicate for image inputs): we expected a free BN win; instead -8 pts to -10 pts Obj F1 throughout training, pixel IoU unchanged. Replicate makes the model believe fields continue past the patch edge and causes adjacent fields to merge into one watershed instance. Zero padding gives a consistent (if artificial) “end-of-field” cue.
- Mixed-country val (austria + cambodia + slovenia): noisier val/loss signal, picked worse held-out checkpoints; -4 pts held-out IoU. We use austria only.

7. Limitations and future work

Coverage. 5,715 (4.1%) of (patch, window) targets never produced a usable URL despite repeated reactivation; another 343 (0.2%) hit hard Planet-side asset failures. These patches still have their original FTW Sentinel-2 imagery but no PlanetScope companion in this release.

Temporal misalignment. Our PlanetScope acquisition date is the closest cloud-free match to the FTW season midpoint, not to the exact Sentinel-2 acquisition date. The gap is usually under two weeks but can

Table 6. Polygon-level evaluation, 11-country held-out, WS+TTA. PQ = SQ×RQ. $AP_{[.5:.95]}$ averages F1 across IoU thresholds. $|\Delta N_{poly}|$ is mean per-patch polygon-count error. Boundary error is mean / p95 symmetric chamfer distance between predicted and GT boundary pixels, in meters (GSD-invariant). Best in bold.

Imagery	Recipe	Panoptic			$AP_{[.5:.95]}$	$ \Delta N_{poly} $	Boundary err (m)	
		PQ	SQ	RQ			mean	p95
S2	augmax B3 (CC-BY)	0.234	0.677	0.303	0.182	65.2	17.01	48.59
S2	augmax B3 (full)	0.246	0.681	0.318	0.191	60.2	16.75	50.13
S2	augmax B7 (CC-BY)	0.248	0.688	0.316	0.196	66.2	16.13	48.53
S2	augmax B7 (full)	0.283	0.750	0.359	0.225	60.2	14.84	43.24
Planet	augmax B3 (CC-BY)	0.273	0.699	0.341	0.223	59.8	8.26	24.14
Planet	augmax B3 + SDF aux (CC-BY)	0.260	0.702	0.326	0.210	62.7	8.21	24.24
Planet	augmax B7 (CC-BY)	0.279	0.700	0.348	0.226	58.2	8.16	24.21
Planet	augmax B3 (full)	0.324	0.710	0.405	0.263	50.9	7.36	22.20

exceed a month in cloudy regions.

Hard transfer regions. Portugal (terraced micro-fields) and Kenya (presence-only labels) collapse to ≤ 0.10 pixel IoU under every configuration we tried. These countries need either targeted data acquisition or a transfer-specific strategy. We do not try to solve them here.

Future work. (i) Rescale the SDF auxiliary target with `resize_aug` so the head doesn't degenerate under aggressive resizing. (ii) Train a B5 row to fill out the backbone $\times$ split grid. (iii) Add 8-band SuperDove samples where they are available, for richer spectral signal.

8. Conclusion

FTW-HD is a 3 m PlanetScope companion to Fields of The World. We did the boring engineering – range-read the COGs, reactivate the broken URLs, rasterize the labels onto the new grid – so that others don't have to. We also showed that a small B3 model trained with the right augmentations does as well as or better than the released 4× larger Sentinel-2 baseline on every metric that does not bake in a pixel-grid bias. Field boundaries are where most of the loss happens, and a 3× finer grid is the cheapest way to localize them.

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